stats101a_hw8

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```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)

install.packages('car', repos = "https://cran.r-project.org")
```

```
##
## The downloaded binary packages are in
##  /var/folders/_9/cmfdrmpd48jdvhyh86gcqxd80000gp/T//RtmpIIn0eD/downloaded_packages
```

```r
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

(1)

```r
df <- data.frame(read.csv('./cars04.csv'))

str(df)
```

```
## 'data.frame':    234 obs. of  13 variables:
##  $ Vehicle.Name        : chr  "Chevrolet Aveo 4dr" "Chevrolet Aveo LS 4dr hatch" "Chevrolet Cavalier
##  $ Hybrid              : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ SuggestedRetailPrice: int  11690 12585 14610 14810 16385 13670 15040 13270 13730 15460 ...
##  $ DealerCost          : int  10965 11802 13697 13884 15357 12849 14086 12482 12906 14496 ...
##  $ EngineSize          : num  1.6 1.6 2.2 2.2 2.2 2 2 2 2 2 ...
##  $ Cylinders           : int  4 4 4 4 4 4 4 4 4 4 ...
##  $ Horsepower          : int  103 103 140 140 140 132 132 130 110 130 ...
##  $ CityMPG             : int  28 28 26 26 26 29 29 26 27 26 ...
##  $ HighwayMPG          : int  34 34 37 37 37 36 36 33 36 33 ...
##  $ Weight              : int  2370 2348 2617 2676 2617 2581 2626 2612 2606 2606 ...
##  $ WheelBase           : int  98 98 104 104 104 105 105 103 103 103 ...
##  $ Length              : int  167 153 183 183 183 174 174 168 168 168 ...
##  $ Width               : int  66 66 69 68 69 67 67 67 67 67 ...
```

```r
numeric_vars <- df[, sapply(df, is.numeric)]
numeric_vars <- numeric_vars[, !names(numeric_vars) %in% c("Hybrid")]


model <- lm(SuggestedRetailPrice ~ ., data = numeric_vars)
summary(model)
```

```
##
## Call:
## lm(formula = SuggestedRetailPrice ~ ., data = numeric_vars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1403.85  -276.86   -55.03   257.55  2584.11
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  349.97628 1461.40052   0.239 0.810953
## DealerCost     1.05418    0.00564 186.923  < 2e-16 ***
## EngineSize   -32.24720  123.05642  -0.262 0.793523
## Cylinders    228.32952   71.99492   3.171 0.001730 **
## Horsepower     2.36212    1.42851   1.654 0.099624 .
## CityMPG      -16.74239   21.46286  -0.780 0.436181
## HighwayMPG    46.75754   24.17910   1.934 0.054403 .
## Weight         0.69920    0.20751   3.370 0.000887 ***
## WheelBase     27.05345   16.36168   1.653 0.099644 .
## Length        -7.32019    7.12296  -1.028 0.305209
## Width        -84.70850   30.21238  -2.804 0.005496 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 532.3 on 223 degrees of freedom
## Multiple R-squared:  0.9989, Adjusted R-squared:  0.9989
## F-statistic: 2.073e+04 on 10 and 223 DF,  p-value: < 2.2e-16
```

(a)

`Vehicle.name` is just the name of the vehicle – it's an identifier similar to ids or keys that provide information to us, the user, but does not provide much information to a model. That's why we exclude it from the model – it does not tell us anything important about `SuggestedRetailPrice` beyond its features / specs that are already extrapolated with the rest of the predictors.

(b)

$\widehat{\text{SuggestedRetailPrice}} = \beta_0+\beta_1(\text{DealerCost})+\beta_2(\text{EngineSize})+\beta_3(\text{Cylinders})+\beta_4(\text{Horsepower})+\beta_5(\text{CityMPG})+\beta_6(\text{HighwayM}$

(c)

```r
summary(model)
```

```
##
## Call:
## lm(formula = SuggestedRetailPrice ~ ., data = numeric_vars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1403.85  -276.86   -55.03   257.55  2584.11
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  349.97628 1461.40052   0.239 0.810953
## DealerCost     1.05418    0.00564 186.923  < 2e-16 ***
## EngineSize   -32.24720  123.05642  -0.262 0.793523
## Cylinders    228.32952   71.99492   3.171 0.001730 **
## Horsepower     2.36212    1.42851   1.654 0.099624 .
## CityMPG      -16.74239   21.46286  -0.780 0.436181
## HighwayMPG    46.75754   24.17910   1.934 0.054403 .
## Weight         0.69920    0.20751   3.370 0.000887 ***
## WheelBase     27.05345   16.36168   1.653 0.099644 .
## Length        -7.32019    7.12296  -1.028 0.305209
## Width        -84.70850   30.21238  -2.804 0.005496 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 532.3 on 223 degrees of freedom
## Multiple R-squared:  0.9989, Adjusted R-squared:  0.9989
## F-statistic: 2.073e+04 on 10 and 223 DF,  p-value: < 2.2e-16
```

slope coefficient: 228.32952 t-value: 3.171 p value: 0.001730

Since the p-value $= 0.001730 < \alpha = 0.05$, there is statistically significant evidence to reject the null hypothesis ($H_0 : B_{\text{cylinders}} = 0$). That is, there is evidence that the number of cylinders has a linear relationship with suggested retail price, controlling for all other predictors.

(d)

For each additional cylinder, the suggested retail price increases by approximately $228.33 on average, all other predictors being the same.

(e)

```r
anova(model)
```

```
## Analysis of Variance Table
##
## Response: SuggestedRetailPrice
##              Df     Sum Sq    Mean Sq    F value    Pr(>F)
## DealerCost    1 5.8714e+10 5.8714e+10 2.0724e+05 < 2.2e-16 ***
## EngineSize    1 7.7453e+06 7.7453e+06 2.7338e+01 3.925e-07 ***
## Cylinders     1 2.7222e+06 2.7222e+06 9.6084e+00  0.002186 **
## Horsepower    1 7.0394e+05 7.0394e+05 2.4847e+00  0.116377
## CityMPG       1 2.1856e+05 2.1856e+05 7.7150e-01  0.380714
## HighwayMPG    1 2.1052e+05 2.1052e+05 7.4310e-01  0.389601
## Weight        1 1.2563e+06 1.2563e+06 4.4344e+00  0.036341 *
## WheelBase     1 3.9621e+04 3.9621e+04 1.3990e-01  0.708785
## Length        1 1.6483e+06 1.6483e+06 5.8179e+00  0.016673 *
## Width         1 2.2271e+06 2.2271e+06 7.8611e+00  0.005496 **
## Residuals   223 6.3178e+07 2.8331e+05
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# get F value: 9.6084e+00
```

```r
# t = sqrt(F)

t <- sqrt(9.6084e+00)

t
```

```
## [1] 3.099742
```

(f)

```r
summary(model)
```

```
##
## Call:
## lm(formula = SuggestedRetailPrice ~ ., data = numeric_vars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1403.85  -276.86   -55.03   257.55  2584.11
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  349.97628 1461.40052   0.239 0.810953
## DealerCost     1.05418    0.00564 186.923  < 2e-16 ***
## EngineSize   -32.24720  123.05642  -0.262 0.793523
```

```
## Cylinders    228.32952   71.99492   3.171 0.001730 **
## Horsepower     2.36212    1.42851   1.654 0.099624 .
## CityMPG      -16.74239   21.46286  -0.780 0.436181
## HighwayMPG    46.75754   24.17910   1.934 0.054403 .
## Weight         0.69920    0.20751   3.370 0.000887 ***
## WheelBase     27.05345   16.36168   1.653 0.099644 .
## Length        -7.32019    7.12296  -1.028 0.305209
## Width        -84.70850   30.21238  -2.804 0.005496 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 532.3 on 223 degrees of freedom
## Multiple R-squared:  0.9989, Adjusted R-squared:  0.9989
## F-statistic: 2.073e+04 on 10 and 223 DF,  p-value: < 2.2e-16
```

F-statistic = 2.073e+04

The F-statistic tests the overall significance of the model, that is:

$H_0 : \beta_1 = \beta_2 = ... = \beta_{10} = 0$

H_a : at least one $\beta_i \neq 0$

Since the p-value < 2.2e-16 < 0.05, we reject H_0 and conclude that at least one predictor has a statistically significant linear relationship with suggested retail price. So the model as a whole is statistically significant.

(2)

```r
df <- data.frame(read.csv('./pgatour2006-3.csv'))

str(df)
```

```
## 'data.frame':    196 obs. of  12 variables:
##  $ Name              : chr  "Aaron Baddeley" "Adam Scott" "Alex Aragon" "Alex Cejka" ...
##  $ TigerWoods        : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ PrizeMoney        : int  60661 262045 3635 17516 16683 107294 50620 57273 86782 23396 ...
##  $ AveDrivingDistance: num  288 301 303 289 288 ...
##  $ DrivingAccuracy   : num  60.7 62 51.1 66.4 63.2 ...
##  $ GIR               : num  58.3 69.1 59.1 67.7 64 ...
##  $ PuttingAverage    : num  1.75 1.77 1.79 1.78 1.76 ...
##  $ BirdieConversion  : num  31.4 30.4 29.9 29.3 29.3 ...
##  $ SandSaves         : num  54.8 53.6 37.9 45.1 52.4 ...
##  $ Scrambling        : num  59.4 57.9 50.8 54.8 57.1 ...
##  $ BounceBack        : num  19.3 19.4 16.8 17.1 18.2 ...
##  $ PuttsPerRound     : num  28 29.3 29.2 29.5 28.9 ...
```

```r
df <- subset(df, select = -c(Name, TigerWoods))

str(df)
```

```
## 'data.frame':    196 obs. of  10 variables:
##  $ PrizeMoney        : int  60661 262045 3635 17516 16683 107294 50620 57273 86782 23396 ...
##  $ AveDrivingDistance: num  288 301 303 289 288 ...
```

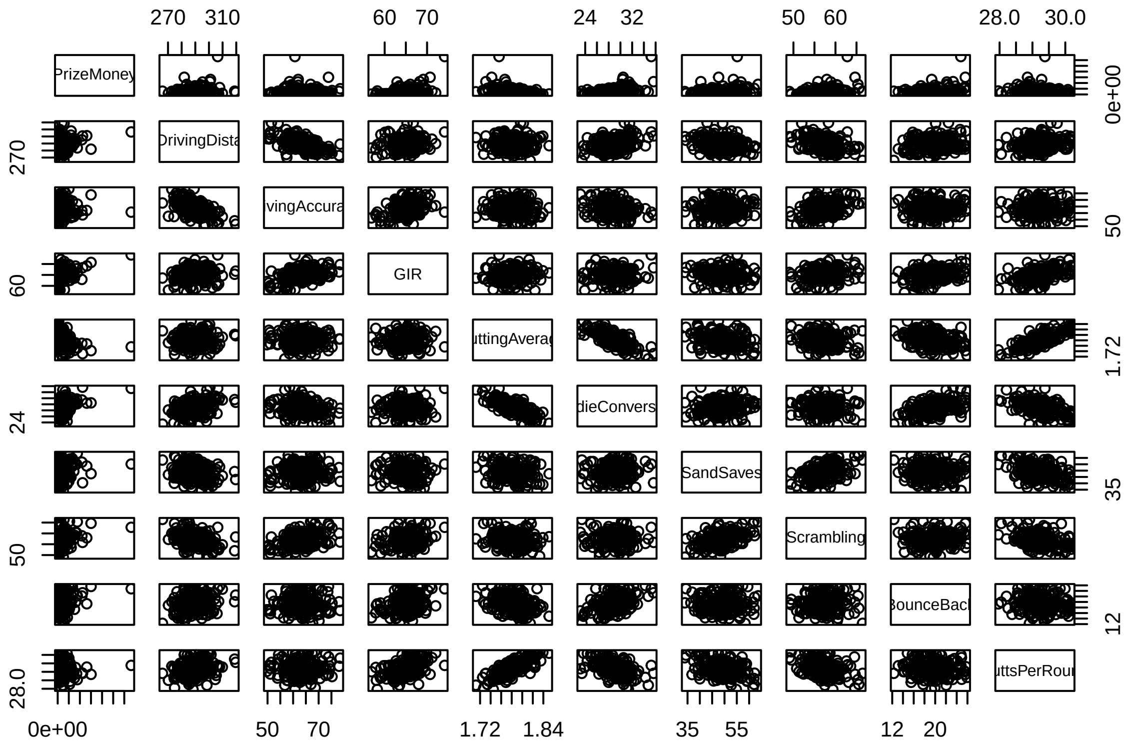
```
##  $ DrivingAccuracy   : num  60.7 62 51.1 66.4 63.2 ...
##  $ GIR               : num  58.3 69.1 59.1 67.7 64 ...
##  $ PuttingAverage    : num  1.75 1.77 1.79 1.78 1.76 ...
##  $ BirdieConversion  : num  31.4 30.4 29.9 29.3 29.3 ...
##  $ SandSaves         : num  54.8 53.6 37.9 45.1 52.4 ...
##  $ Scrambling        : num  59.4 57.9 50.8 54.8 57.1 ...
##  $ BounceBack        : num  19.3 19.4 16.8 17.1 18.2 ...
##  $ PuttsPerRound     : num  28 29.3 29.2 29.5 28.9 ...
```

```r
model <- lm(PrizeMoney ~ ., data = df)
summary(model)
```

```
##
## Call:
## lm(formula = PrizeMoney ~ ., data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -80475 -26186  -6671  15209 417966
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -1174845.3   591253.0  -1.987  0.04839 *
## AveDrivingDistance     -720.8      766.0  -0.941  0.34795
## DrivingAccuracy       -2457.9     1104.0  -2.226  0.02720 *
## GIR                   10709.3     3761.8   2.847  0.00491 **
## PuttingAverage       123072.1   579670.7   0.212  0.83209
## BirdieConversion      11758.4     3801.9   3.093  0.00229 **
## SandSaves              1074.7      759.4   1.415  0.15868
## Scrambling             4313.5     2477.8   1.741  0.08336 .
## BounceBack              568.5     1585.2   0.359  0.72028
## PuttsPerRound           701.0    37306.6   0.019  0.98503
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 50270 on 186 degrees of freedom
## Multiple R-squared:  0.4097, Adjusted R-squared:  0.3811
## F-statistic: 14.34 on 9 and 186 DF,  p-value: < 2.2e-16
```

(a)

```r
pairs(df)
```



PuttingAverage and PuttsPerRound.

DrivingDistance and DrivingAccuracy

PuttingAverage and BirdieConversion

(b)

```r
vif(model)[vif(model) > 5]
```

```
##              GIR   PuttingAverage BirdieConversion    PuttsPerRound
##         8.092384        15.853659         5.430257        20.951499
```

PuttingAverage, BirdieConversion, and PuttsPerRound are greater than 5, which relates to the issues with collinearity that I saw in (a). Interestingly, and even after investigating the scatterplots in hindsight, GIR has a VIF value > 5 but I don't really think the collinearity is too big with it.

(c)

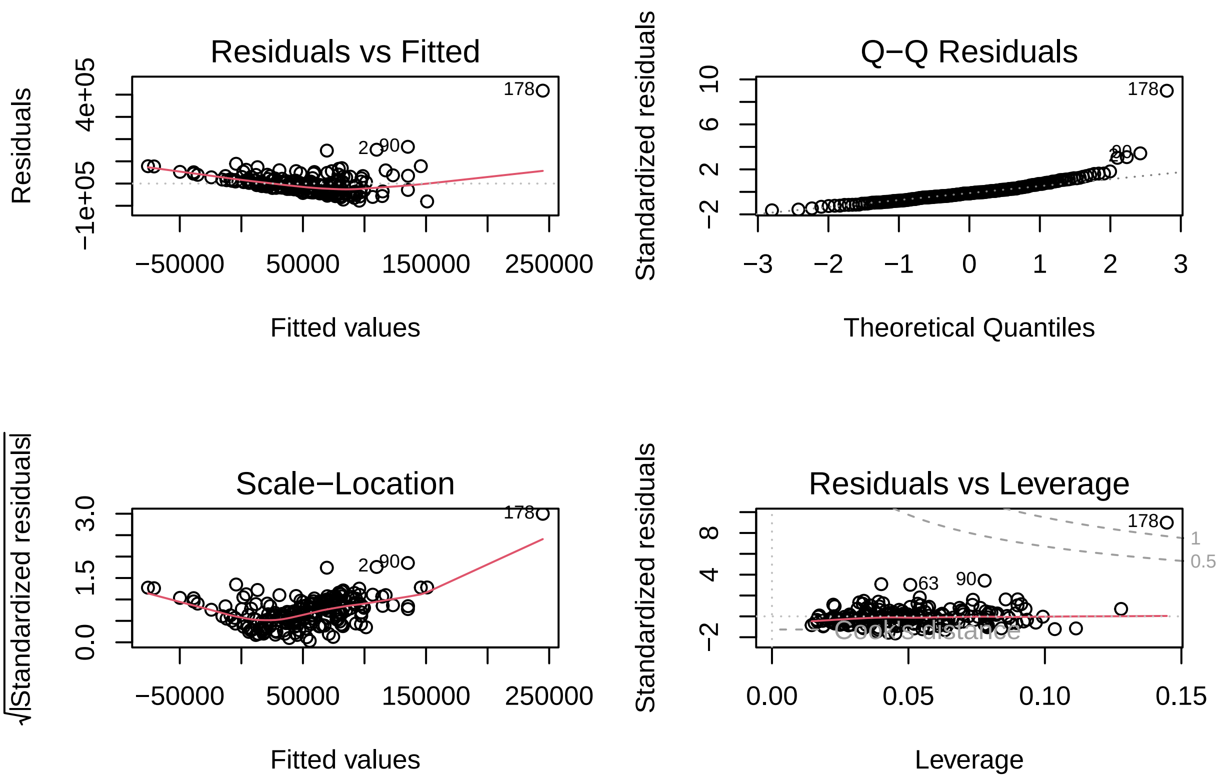
An inflated standard error directly inflates the denominator of the t-statistic since

$$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

So a larger Standard Error results in a smaller magnitude t score which results in a larger p-value. This means the variable may appear statistically insignificant not because it truly has no effect, but because collinearity is masking it.

(d)

```r
par(mfrow = c(2,2))
plot(model)
```



Non-constant variance: - residuals v fitted plot show a fan pattern where variance increases with fitted values - scale-location plot has strong upward trend

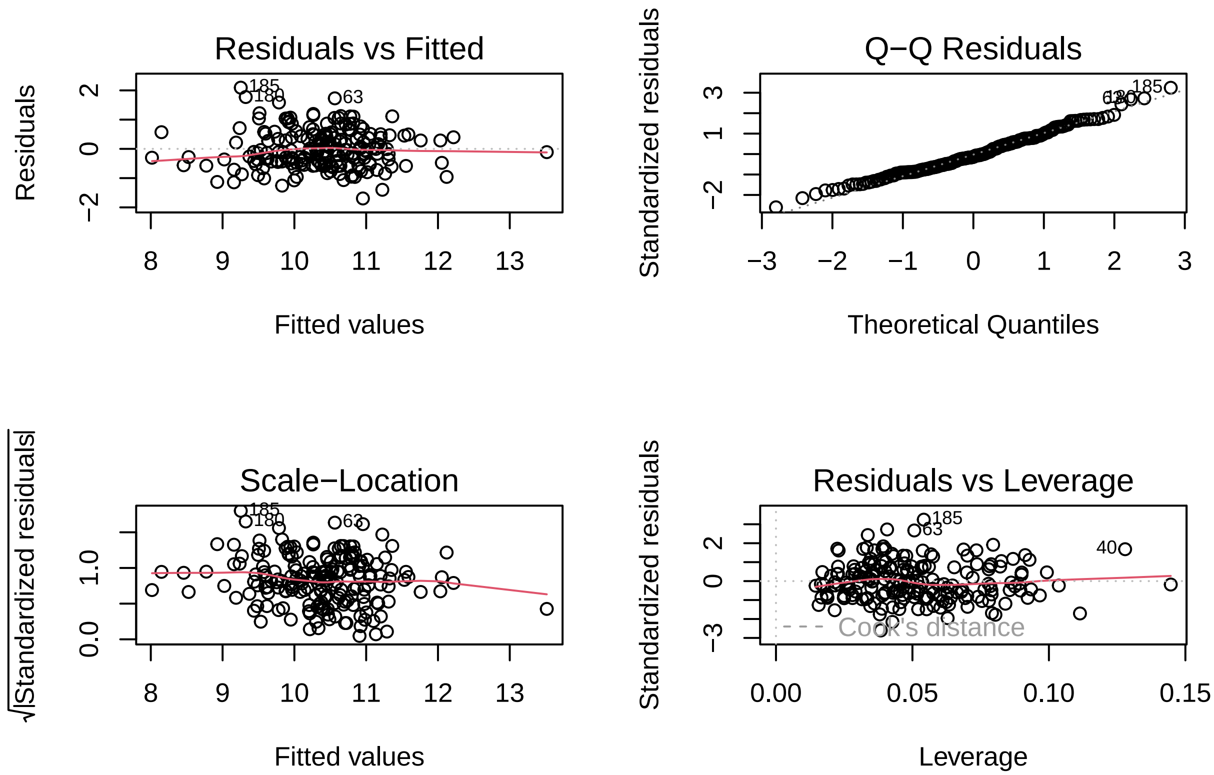
non-normal: - QQplot has significant deviation from the straight line at the upper tail.

trend/non-linearity: - residuals vs fitted plot shows a clear pattern/trend where it decreases in a straight line then goes back up .

(e)

```r
log_model <- lm(log(PrizeMoney) ~ ., data = df)

par(mfrow = c(2,2))
plot(log_model)
```



Yes a log transform of the response variable makes the validity much better.

The constant variance is much better supported. The QQplot is much more following hte line. Lack of a trend in residuals v. fitted as well as pretty constant variacnce.

(f)

```
n <- nrow(df)
p <- 9
threshold <- 2*(p+1)/n
threshold
```

```
## [1] 0.1020408
```

```
# get leverage values
leverage <- hatvalues(model)
which(leverage > threshold)
```

```
##  28  40 168 178
##  28  40 168 178
```

potential bad high leverage points = observations 28 40 168 178

(g)

```r
cooks <- cooks.distance(model)
which(cooks > 1)
```

```
## 178
## 178
```

observation 178 is a high influence point.